



# Cambridge (CIE) IGCSE Chemistry



Your notes

## Giant Structures

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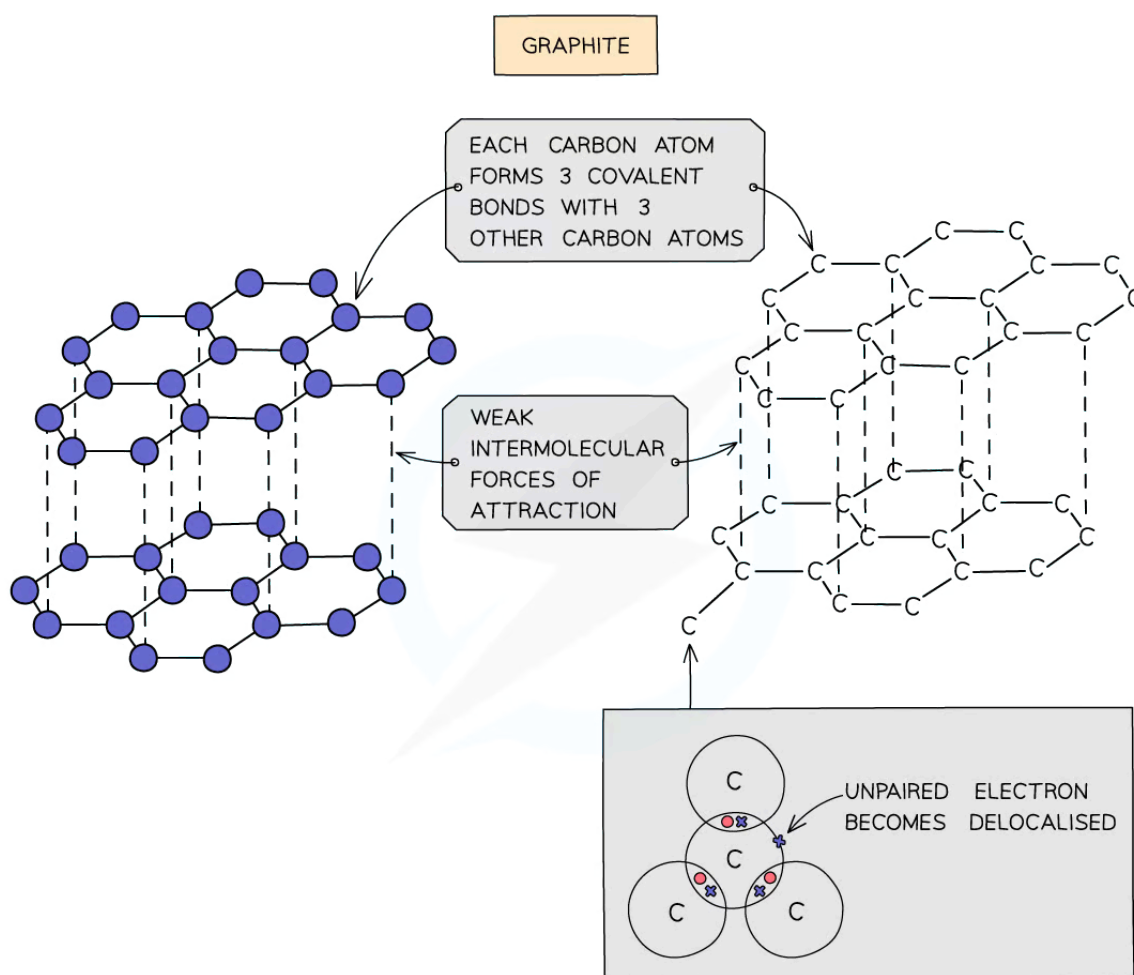
## Diamond & Graphite

# Structure of graphite & diamond

## What is the structure of graphite?

- Each carbon atom in graphite is bonded to **three** others forming **layers of hexagons**, leaving one free electron per carbon atom which becomes delocalised
- The covalent bonds within the layers are very strong, but the layers are attracted to each other by weak **intermolecular forces**

## Diagram to show the bonding and structure in graphite



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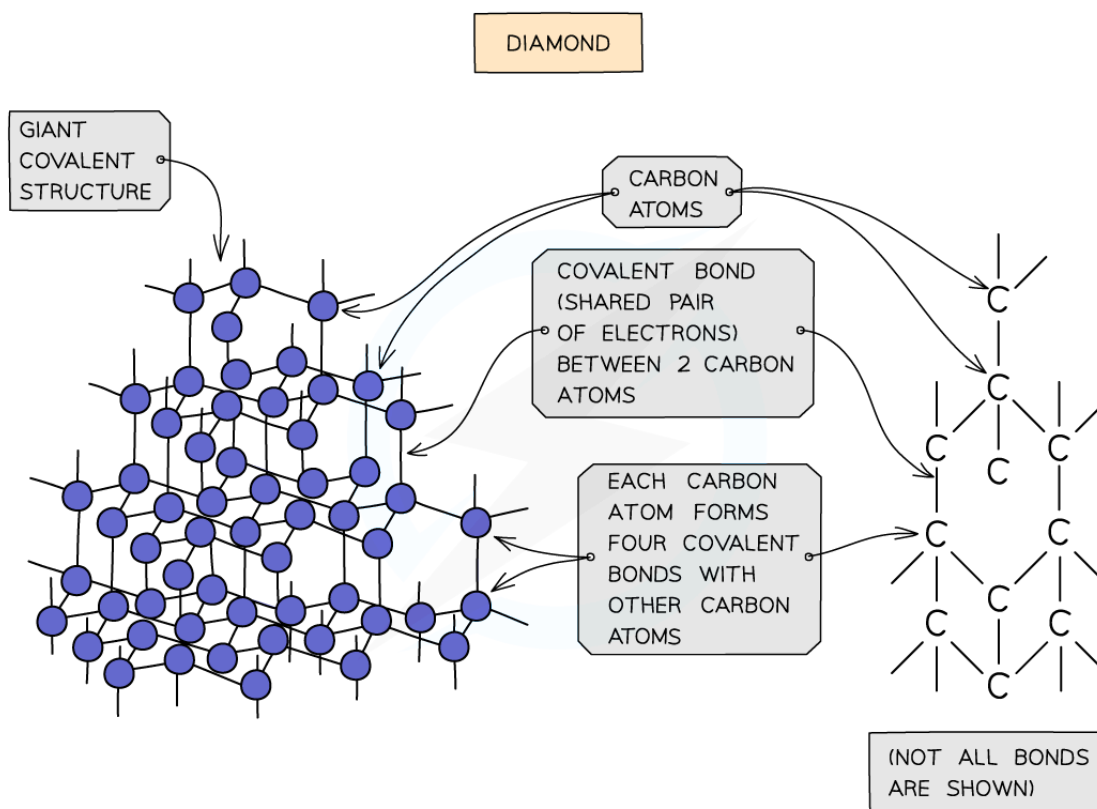
### The structure and bonding in graphite

- Diamond and graphite are **allotropes** of carbon which have **giant covalent structures**
- Both substances contain only carbon atoms but due to the differences in bonding arrangements they are physically completely different
- Giant covalent structures contain billions of non-metal atoms, each joined to adjacent atoms by covalent bonds forming a giant lattice structure

## What is the structure of diamond?

- In diamond, each carbon atom bonds with four other carbons, forming a **tetrahedron**
- All the covalent bonds are identical, very strong and there are no **intermolecular forces**

### Diagram to show the formation of a covalent bond



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Diagram showing the structure and bonding arrangement in diamond



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## Uses of graphite & diamond

### What are the properties and uses of graphite?

- Graphite **conducts electricity**
  - Each carbon atom is bonded to three others leaving one free electron per carbon atom
  - These free (delocalised) electrons exist in between the layers
  - They are free to move through the structure and carry charge
- Graphite has a **high melting point**
  - Graphite has a giant covalent structure
  - There are strong covalent bonds between the carbon atoms
  - These need lots of energy to break
- Graphite is **slippery**
  - Graphite is arranged in layers
  - Although the atoms within the layers are joined by strong covalent bonds, the layers have only weak intermolecular forces between them
  - As a result the layers can slide over each other
  - This property allows graphite to be used in pencils and as an industrial lubricant



#### Examiner Tips and Tricks

Don't confuse pencil lead with the metal lead – they have nothing in common. Pencil lead is actually graphite, and historical research suggests that in the past, lead miners sometimes confused the mineral galena (lead sulfide) with graphite; since the two looked similar they termed both minerals 'lead'. The word graphite derives from the Latin word 'grapho' meaning 'I write', so it is a well named mineral!

### What are the properties and uses of diamond?

- Diamond does **not conduct electricity**
  - All the outer shell electrons in carbon are held in the four covalent bonds around each carbon atom
  - As a result, there are no freely moving particles to carry a charge



Your notes

- Diamond has a very **high melting point**
  - Diamond has a giant covalent structure
  - There are strong covalent bonds between the carbon atoms
  - These need lots of energy to break
- It is extremely **hard** and **dense**
  - It has strong covalent bonds and each carbon atom is bonded to four other carbon atoms
  - Diamond's hardness makes it very useful in cutting tools like drills
- Diamond has the following physical properties:
  - It does not conduct electricity
  - It has a very high melting point
  - It is extremely hard and dense



### Examiner Tips and Tricks

Diamond is the hardest naturally occurring mineral, but it is by no means the strongest. Students often confuse **hard** with **strong**, thinking it is the opposites of weak. Diamonds are hard, but brittle – that is, they can be smashed fairly easily with a hammer. The opposite of saying a material is hard is to describe it as **soft**.



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## Silicon(IV) Oxide

### Structure of silicon(IV) oxide

#### What is the structure of silicon(IV) oxide?

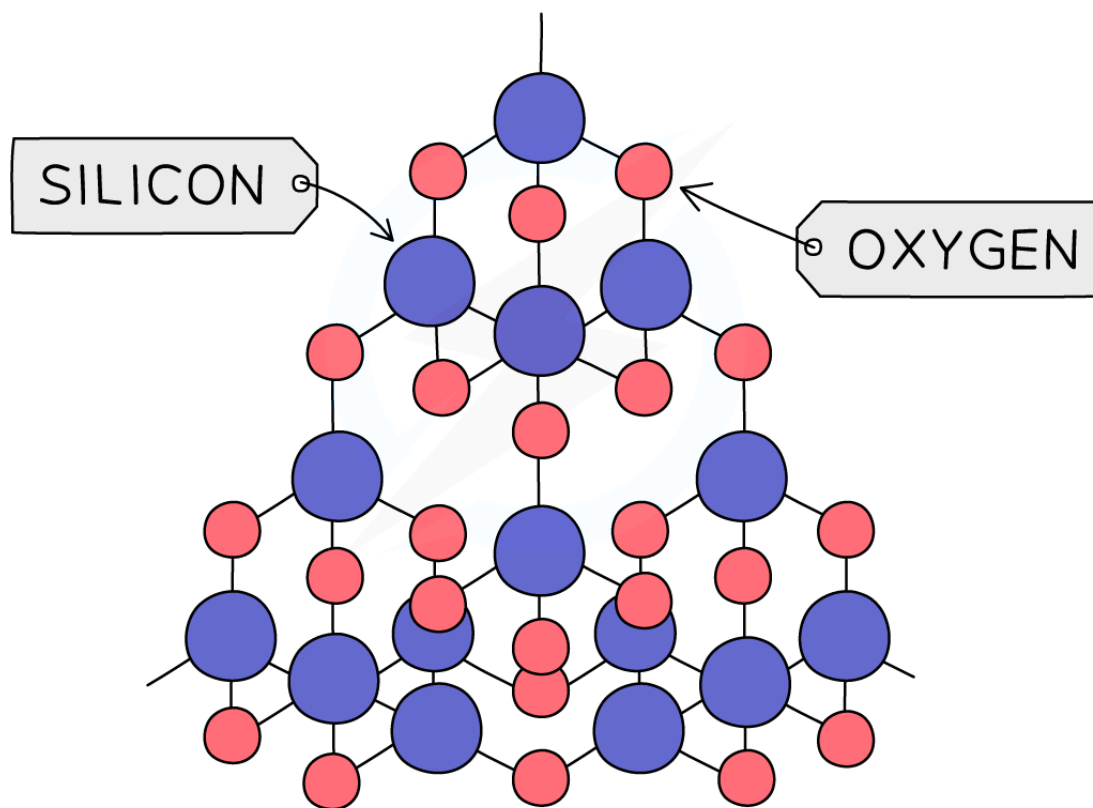
##### Extended tier only

- Silicon(IV) oxide (also known as silicon dioxide or silica),  $\text{SiO}_2$ , is a macromolecular compound which occurs naturally as **sand** and **quartz**
- Each oxygen atom forms covalent bonds with **2** silicon atoms and each silicon atom in turn forms covalent bonds with **4** oxygen atoms
- A tetrahedron is formed with one silicon atom and four oxygen atoms, similar to diamond

#### The structure of silicon(IV) oxide



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Diagram showing the structure of  $\text{SiO}_2$  with the silicon atoms in blue and the oxygen atoms in red

## Comparing diamond & silicon(IV) oxide

### How does diamond compare to silicon(IV) oxide?

#### Extended tier only

- $\text{SiO}_2$  has lots of very strong covalent bonds and no intermolecular forces so it has similar properties to diamond
- It is very **hard**, has a very **high** boiling point, is insoluble in water and does not conduct electricity

- $\text{SiO}_2$  is cheap since it is available naturally and is used to make sandpaper and to line the inside of furnaces



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## Metallic Bonding

# Metallic bonding

## The structure of a metal

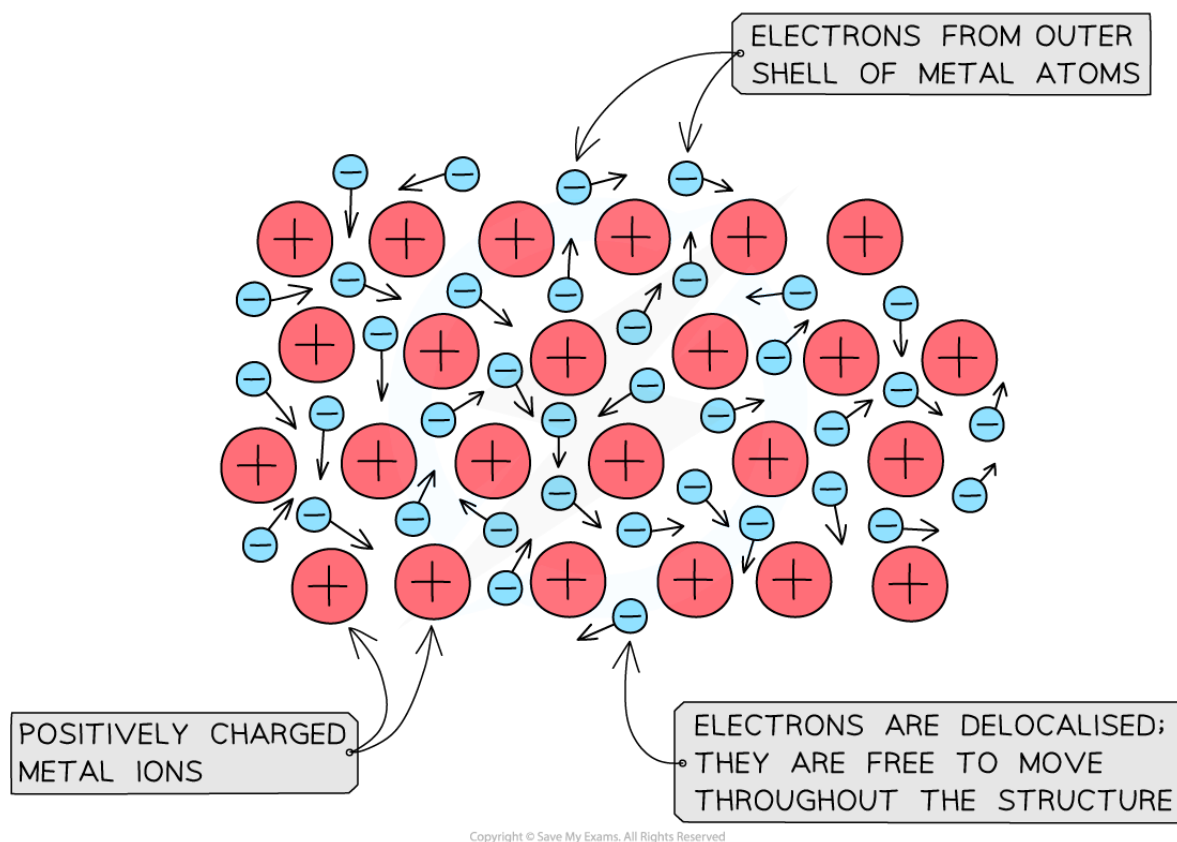
### Extended tier only

- Metals consist of **giant** structures
- Within the metal lattice, the atoms lose their outer electrons and become positively charged metal **ions**
  - The outer electrons no longer belong to any specific metal atom and are said to be **delocalised**
  - This means they can move freely between the positive metal ions and act like a “*sea of electrons*”
- The metallic bond is the strong force of **attraction** between the positive metal ions and the delocalised electrons
- This type of bonding occurs in metals and metal alloys, which are mixtures of metal

### Diagram to show metallic bonding



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*Diagram showing metallic lattice structure with delocalised electrons*

## Properties of metals

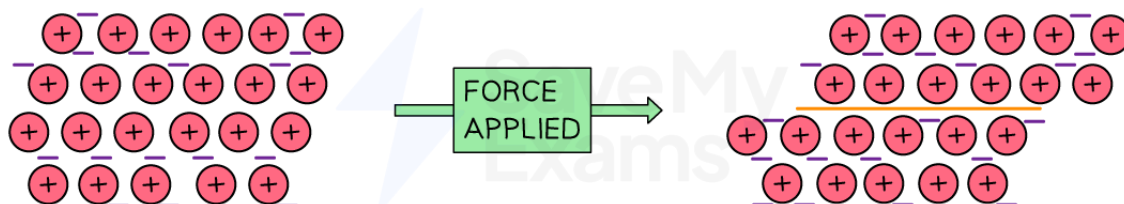
### What are the properties of metals?

#### Extended tier only

- Most metals have **high melting and boiling points**
  - There are strong electrostatic forces of attraction between the positive metal ions and the negative delocalised electrons within the metal lattice structure
  - These needs lots of energy to be broken
- Metals are **good conductors of heat and electricity**
  - The delocalised electrons are free to move and carry a charge through the whole structure
- Most metals are **malleable**

- This means they can be hammered into shape
- This is because the atoms are arranged in layers which can slide over each when force is applied

## Malleability of metals



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*When a force is applied, the layers of positive ions slide over each other*

- Most metals are **ductile**
  - This means they can be pulled into wires
  - This is also because the atoms are arranged in layers which can slide over each when force is applied



### Examiner Tips and Tricks

When explaining why metals can conduct electricity, be careful of the terminology you use. Don't get confused with ionic compounds.

**Metals** can conduct electricity as they have free **electrons** that can carry charge whereas molten or aqueous **ionic** compounds can conduct electricity because they have free **ions** that can carry charge.



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